

Question			Marking Details	Marks Available					
				AO1	AO2	AO3	TOTAL	Maths	Prac
5	(a)	i	Intraspecific (1) Space/ food (1)	1	1		2		
		ii	<i>Semibalanus</i> could not survive in an area that experienced so much desiccation / extreme changes/ OWTTE Reject cannot survive at high tide level not qualified			1	1		
		iii	<i>Semibalanus</i> is a more successful competitor in the lower zone/ <i>Chthamalus</i> is outcompeted in this area (1) There is <u>interspecific</u> competition normally/ A lack of <u>interspecific</u> competition after clearing (1)		2		2		
	(b)	i	A. Choose {same / random} {positions/ aspect} on each statue (1) B. Use of quadrat to count {number/ percentage cover} of barnacles (1) C. Repeat 1) D. Calculate a mean (1)	3	1		4		3
		ii	Due to water potential in the cell being <u>lower</u> than the surrounding water (1) water would <u>move in by osmosis</u> (1) cells would lyse/ burst (1)			3	3		
	(c)	i	{Genetic/ DNA} {fingerprint/ profiling}/ DNA base sequencing/ DNA hybridisation	1					
		ii	2 correct answers for 1 mark from: UV light (Chemical) mutagen Ionizing radiation Errors in DNA replication NOT radiation alone	1					

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	(d)		Any four × (1) from: A. Could outcompete {native/ other} species (1) B. May not have a natural predator in this area (1) C. This will reduce the biodiversity (around the coast)/ impact on food chains (1) D. Could cause damage to the {harbours/ boats} (more friction so more fuel costs)/ boats use more fuel so higher economic impact (1) E. Economic impact of removal of barnacles from {boats/ harbour}/ prevention of barnacle growth (1) F. Introduction of disease (1)			4	4		
			QUESTION 5 TOTAL	6	4	8	18	0	3